

Maiken Høgh Mikkelsen

Professor of Nanophotonics

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Experimental nanophotonics, quantum and nonlinear optics, and metasurfaces. Research focuses on controlling light–matter interactions in nanogap cavities and low-dimensional materials, with applications in ultrafast optoelectronics, quantum technologies, and biosensing.

ACADEMIC APPOINTMENTS

2026–present	Professor of Nanophotonics, Niels Bohr Institute, University of Copenhagen (since July 2026).
2025–2026	Professor, Duke University (through June 2026).
2018–2025	James N. and Elizabeth H. Barton Associate Professor, Duke University.
2012–2017	Assistant Professor, Electrical and Computer Engineering and Physics, Duke University; Nortel Networks Assistant Professor, 2015–2017.
2018–2019	Visiting Associate Professor, Materials Science and Engineering, Stanford University.
2010–2012	Postdoctoral Fellow in Nanophotonics, University of California, Berkeley.

EDUCATION

2009	Ph.D. in Physics, University of California, Santa Barbara.
2007	M.A. in Physics, University of California, Santa Barbara.
2004	B.S. in Physics, University of Copenhagen.

SELECTED HONORS AND FELLOWSHIPS

2025	Fellow, Optica
2021	Stansell Family Distinguished Research Award, Duke University
2020	ACS Photonics Young Investigator Award
2019	Moore Inventor Fellow, Gordon and Betty Moore Foundation
2017	Maria Goeppert Mayer Award, American Physical Society
2017	Early Career Achievement Award, SPIE
2016	Cottrell Scholar Award and Scialog Fellowship, Research Corporation for Science Advancement
2014	Ralph E. Powe Junior Faculty Award
2011	Ph.D. Thesis Prize in Quantum Electronics and Optics, European Physical Society

SELECTED RESEARCH CONTRIBUTIONS

Ultrafast single-photon sources at telecommunications wavelengths. Integrated colloidal quantum dots into plasmonic nanogap cavities to achieve room-temperature single-photon emission at 1550 nm, with spontaneous-emission-rate enhancements of approximately 10,000 and detected count rates of 12.6 MHz at 1550 nm and 15 MHz at 1350 nm. ACS Nano 19, 19035–19045 (2025).

Nanophotonic control of thermal photodetection. Combined metasurface absorption and spectral filtering with a 100 nm pyroelectric layer to demonstrate thermal photodetection with response times below 700 ps. Subsequent work demonstrated metasurface-enhanced thermal photodetection at gigahertz frequencies. Nature Materials 19, 158–162 (2020); Advanced Functional Materials 36, 2420953 (2026).

Plasmonic enhancement of diagnostic immunoassays. Integrated a sandwich immunoassay microarray within plasmonic nanogap cavities, increasing fluorescence 151-fold and improving the detection limit for B-type natriuretic peptide 14-fold. Nano Letters 20, 4330–4336 (2020).

Spontaneous-emission control in nanogap cavities. Elucidated the mechanisms of large Purcell enhancement and demonstrated approximately 1,000-fold enhancement of spontaneous-emission rates in dye molecules and semiconductor quantum dots, followed by ultrafast single-photon emission from an individual quantum dot. Nature Photonics 8, 835 (2014); Nature Communications 6, 7788 (2015); Nano Letters 16, 270 (2016).

Optical measurement and control of individual electron spins. Contributed to nondestructive optical measurement, observation of coherent spin dynamics, and picosecond coherent optical manipulation of a single electron spin in a quantum dot. Science 314, 1916 (2006); Nature Physics 3, 770 (2007); Science 320, 349–352 (2008).

SELECTED RESEARCH FUNDING

2021	Multidisciplinary University Research Initiative (MURI) Center Award, Air Force Office of Scientific Research, U.S. Department of Defense. Award: US\$7.5 million.
2019	R01 research award, National Institutes of Health. Award: US\$3.4 million.
2017	Young Investigator Program Award, Office of Naval Research.
2016	Young Investigator Program Award, Army Research Office.
2015	CAREER Award, National Science Foundation.
2015	Young Investigator Program Award, Air Force Office of Scientific Research.

JOURNAL PUBLICATIONS

† Corresponding author * Equal contribution

52. D. Acil, H. Li & **M. H. Mikkelsen**[†]. Thin Film Interference in Diamond Membranes for Control of Silicon Vacancy Center Emission. *Advanced Optical Materials* **14**, e01609 (2026). *Front cover*.
51. E. Shin, R. E. Bangle, N. C. Wilson, S. B. Nikodemski, J. H. Vella & **M. H. Mikkelsen**[†]. Metasurface-Enhanced Thermal Photodetector Operating at Gigahertz Frequencies. *Advanced Functional Materials* **36**, 2420953 (2026).
50. E. Shin, R. E. Bangle & **M. H. Mikkelsen**[†]. Real-Time Tuning of Plasmonic Nanogap Cavity Resonances through Solvent Environments. *Nanophotonics* **14**, 3889–3895 (2025).
49. S. Zhang, A. J. Traverso, E. A. Dolgoplova, A. Singh, H. Kishida, M. Y. Livshits, C. J. Sheehan, E. G. Bowes, J. A. Hollingsworth & **M. H. Mikkelsen**[†]. Solution-processed ultrafast, room-temperature single-photon source at 1550 nm. *ACS Nano* **19**, 19035–19045 (2025).
48. H. Li, D. Acil, A. Boyce, C. Pederson, S. Chakravarthi, N. Yama, K.-M. C. Fu & **M. H. Mikkelsen**[†]. Room-temperature Picosecond Single Photon Emission from Silicon Vacancy Center in Diamond. *ACS Nano* **19**, 19310–19317 (2025). *Cover article*.
47. A. M. Boyce, H. Li, N. C. Wilson, D. Acil, A. Shams-Ansari, S. Chakravarthi, C. Pederson, Q. Shen, N. Yama, K.-M. C. Fu, M. Loncar & **M. H. Mikkelsen**[†]. Plasmonic Diamond Membranes for Ultrafast Silicon Vacancy Emission. *Nano Letters* **24**, 3575–3580 (2024).
46. R. E. Bangle & **M. H. Mikkelsen**[†]. Tracking light-induced charge transport. *Science* **382**, 264–265 (2023). *Perspective*.
45. N. C. Wilson, E. Shin, R. E. Bangle, S. B. Nikodemski, J. H. Vella & **M. H. Mikkelsen**[†]. Ultrathin Pyroelectric Photodetector with Integrated Polarization-Sensing Metasurface. *Nano Letters* **23**, 8547–8552 (2023).
44. R. E. Bangle, H. Li & **M. H. Mikkelsen**[†]. Uncovering the Mechanisms of Triplet-Triplet Annihilation Upconversion Enhancement via Plasmonic Nanocavity Tuning. *ACS Nano* **17**, 24022–24032 (2023).
43. D. Semeniak, D. F. Cruz, A. Chilkoti & **M. H. Mikkelsen**[†]. Plasmonic Fluorescence Enhancement in Diagnostics for Clinical Tests at Point-of-Care: A Review of Recent Technologies. *Advanced Materials* **35**, 2107986 (2023). *Review*.
42. J. W. Stewart, T. Nebabu & **M. H. Mikkelsen**[†]. Control of Nanoscale Heat Generation with Lithography-Free Metasurface Absorbers. *Nano Letters* **22**, 5151–5157 (2022).
41. A. M. Boyce, J. W. Stewart, J. Avila, Q. Shen, S. Zhang, V. D. Wheeler & **M. H. Mikkelsen**[†]. Actively Tunable Metasurfaces via Plasmonic Nanogap Cavities with sub-10 nm VO₂ Films. *Nano Letters* **22**, 3525–3531 (2022).

40. H. Kishida & **M. H. Mikkelsen**[†]. Ultrafast Lifetime and Bright Emission from Graphene Quantum Dots Using Plasmonic Nanogap Cavities. *Nano Letters* **22**, 904–910 (2022).
39. A. J. Traverso^{*}, J. Huang^{*}, T. Peyronel, G. Yang, T. G. Tiecke & **M. H. Mikkelsen**[†]. Low-loss, centimeter-scale plasmonic metasurface for ultrafast optoelectronics. *Optica* **8**, 202–207 (2021).
38. C. M. Fontes, B. D. Lipes, J. Liu, K. N. Agans, A. Yan, P. Shi, D. F. Cruz, G. Kelly, K. M. Luginbuhl, D. Y. Joh, S. L. Foster, J. Heggstad, A. Hucknall, **M. H. Mikkelsen**, C. F. Pieper, R. W. Horstmeyer, T. W. Geisbert, M. D. Gunn & A. Chilkoti. Ultrasensitive Point-of-Care Immunoassay for Secreted Glycoprotein Detects Ebola Infection Earlier than PCR. *Science Translational Medicine* **13** (588) (2021).
37. Q. Shen, A. Shams-Ansari, A. M. Boyce, N. C. Wilson, T. Cai, M. Loncar & **M. H. Mikkelsen**[†]. A metasurface-based diamond frequency converter using plasmonic nanogap resonators. *Nanophotonics* **10**, 589–595 (2021).
36. J. W. Stewart, N. C. Wilson & **M. H. Mikkelsen**[†]. Nanophotonic Engineering: A New Paradigm for Spectrally Sensitive Thermal Photodetectors. *ACS Photonics* **8**, 71–84 (2021).
35. Q. Shen, W. Jin, G. Yang, A. W. Rodriguez & **M. H. Mikkelsen**[†]. Active control of multiple, simultaneous nonlinear optical processes in plasmonic nanogap cavities. *ACS Photonics* **7**, 901–907 (2020).
34. D. F. Cruz, C. M. Fontes, D. Semeniak, J. Huang, A. Hucknall, A. Chilkoti & **M. H. Mikkelsen**[†]. Ultrabright Fluorescence Readout of a Point-of-Care Immunoassay using Plasmonic Metasurfaces. *Nano Letters* **20**, 4330–4336 (2020).
33. J. W. Stewart, J. H. Vella, W. Li, S. Fan & **M. H. Mikkelsen**[†]. Ultrafast pyroelectric photodetection with on-chip spectral filters. *Nature Materials* **19**, 158–162 (2020).
32. G. Yang, Q. Shen, Y. Niu, H. Wei, B. Bai[†], **M. H. Mikkelsen**[†] & H.-B. Sun. Unidirectional, Ultrafast, and Bright Spontaneous Emission Source Enabled by a Hybrid Plasmonic Nanoantenna. *Laser & Photonics Reviews*, 1900213 (2020).
31. Q. Shen, A. M. Boyce, G. Yang & **M. H. Mikkelsen**[†]. Polarization-Controlled Nanogap Cavity with Dual-Band and Spatially Overlapped Resonances. *ACS Photonics* **6**, 1916–1921 (2019).
30. J. J. Baumberg, J. Aizpurua, **M. H. Mikkelsen** & D. R. Smith. Extreme nanophotonics from ultrathin metallic gaps. *Nature Materials* **18**, 668–678 (2019).
29. J. Huang^{*}, A. J. Traverso^{*}, G. Yang & **M. H. Mikkelsen**[†]. Real-Time Tunable Strong Coupling: From Individual Nanocavities to Metasurfaces. *ACS Photonics* **6**, 838–843 (2019).
28. W. M. Wilson, J. W. Stewart & **M. H. Mikkelsen**[†]. Surpassing Single Line Width Active Tuning with Photochromic Molecules Coupled to Plasmonic Nanoantennas. *Nano Letters* **18**, 853–858 (2018).

27. J. Huang, G. M. Akselrod, T. Ming, J. Kong & **M. H. Mikkelsen**[†]. Tailored emission spectrum of 2D semiconductors using plasmonic nanocavities. *ACS Photonics* **5**, 552–558 (2018). *ACS Editors' Choice*.
26. Q. Shen, T. B. Hoang, G. Yang, V. D. Wheeler & **M. H. Mikkelsen**[†]. Probing the origin of highly-efficient third harmonic generation in plasmonic nanogaps. *Optics Express* **26**, 20718–20725 (2018).
25. T. B. Hoang, G. M. Akselrod, A. Yang, T. W. Odom & **M. H. Mikkelsen**[†]. Millimeter-Scale Spatial Coherence from a Plasmon Laser. *Nano Letters* **17**, 6690–6695 (2017).
24. M. E. Sykes, J. W. Stewart, G. M. Akselrod, X.-T. Kong, Z. Wang, D. J. Gosztola, A. B. F. Martinson, D. Rosenmann, **M. H. Mikkelsen**[†], A. O. Govorov[†] & G. P. Wiederrecht[†]. Enhanced generation and anisotropic Coulomb scattering of hot electrons in an ultra-broadband plasmonic nanopatch metasurface. *Nature Communications* **8**, 986 (2017).
23. J. Huang, T. B. Hoang, T. Ming, J. Kong & **M. H. Mikkelsen**[†]. Temporal and spatial valley dynamics in two-dimensional semiconductors probed via Kerr rotation. *Physical Review B* **95**, 075428 (2017).
22. J. W. Stewart, G. M. Akselrod, D. R. Smith & **M. H. Mikkelsen**[†]. Toward Multispectral Imaging with Colloidal Metasurface Pixels. *Advanced Materials* **29**, 1602971 (2017). *Back cover*.
21. G. M. Akselrod, M. C. Weidman, Y. Li, C. Argyropoulos, W. A. Tisdale & **M. H. Mikkelsen**[†]. Efficient nanosecond photoluminescence from infrared PbS quantum dots coupled to plasmonic nanoantennas. *ACS Photonics* **3**, 1741–1746 (2016).
20. T. B. Hoang^{*}, G. M. Akselrod^{*} & **M. H. Mikkelsen**[†]. Ultrafast room-temperature single photon emission from quantum dots coupled to plasmonic nanocavities. *Nano Letters* **16**, 270 (2016).
19. T. B. Hoang & **M. H. Mikkelsen**[†]. Broad electrical tuning of plasmonic nanoantennas at visible frequencies. *Applied Physics Letters* **108**, 183107 (2016).
18. W. Ge, T. B. Hoang, **M. H. Mikkelsen** & A. D. Stiff-Roberts. RIR MAPLE deposition of plasmonic silver nanoparticles. *Applied Physics A* **122**, 824 (2016).
17. J. Huang, T. B. Hoang & **M. H. Mikkelsen**[†]. Probing the origin of excitonic states in monolayer WSe₂. *Scientific Reports* **6**, 22414 (2016).
16. T. B. Hoang, J. Huang & **M. H. Mikkelsen**[†]. Colloidal synthesis of nanopatch antennas for applications in plasmonics and nanophotonics. *Journal of Visualized Experiments* **111**, e53876 (2016).
15. G. M. Akselrod, J. Huang, T. B. Hoang, P. T. Bowen, L. Su, D. R. Smith & **M. H. Mikkelsen**[†]. Large-area metasurface perfect absorbers from visible to near infrared. *Advanced Materials* **27**, 7897 (2015). *Front cover*.

14. T. B. Hoang*, G. M. Akselrod*, C. Argyropoulos, J. Huang, D. R. Smith & **M. H. Mikkelsen**[†]. Ultrafast spontaneous emission source using plasmonic nanoantennas. *Nature Communications* **6**, 7788 (2015).
13. G. M. Akselrod, T. Ming, C. Argyropoulos, T. B. Hoang, Y. Lin, X. Ling, D. R. Smith, J. Kong & **M. H. Mikkelsen**[†]. Leveraging Nanocavity Harmonics for Control of Optical Processes in 2D Semiconductors. *Nano Letters* **15**, 3578 (2015).
12. A. Baron, T. B. Hoang, C. Fang, **M. H. Mikkelsen** & D. R. Smith. Ultrafast self-action of surface plasmon polaritons at an air/metal interface. *Physical Review B* **91**, 195412 (2015).
11. A. Yang, T. B. Hoang, M. Dridi, C. Deeb, **M. H. Mikkelsen**, G. C. Schatz & T. W. Odom. Real-time Tunable Lasing from Plasmonic Nanocavity Arrays. *Nature Communications* **6**, 6939 (2015).
10. P. Kolchin*, N. Pholchai*, **M. H. Mikkelsen***, J. Oh, S. Ota, M. S. Islam, X. Yin & X. Zhang. High Purcell Factor Due to Coupling of a Single Emitter to a Dielectric Slot Waveguide. *Nano Letters* **15**, 464 (2015).
9. G. M. Akselrod, C. Argyropoulos, T. B. Hoang, C. Ciraci, C. Fang, J. Huang, D. R. Smith & **M. H. Mikkelsen**[†]. Probing the mechanisms of large Purcell enhancement in plasmonic nanoantennas. *Nature Photonics* **8**, 835 (2014).
8. A. Rose, T. B. Hoang, F. McGuire, J. J. Mock, C. Ciraci, D. R. Smith & **M. H. Mikkelsen**[†]. Control of radiative processes using tunable plasmonic nanopatch antennas. *Nano Letters* **14**, 4797 (2014).
7. J. B. Lassiter, X. Chen, X. Liu, C. Ciraci, T. B. Hoang, S. Larouche, S.-H. Oh, **M. H. Mikkelsen** & D. R. Smith. Third-Harmonic Generation Enhancement by Film-Coupled Plasmonic Stripe Resonators. *ACS Photonics* **1**, 1212 (2014).
6. T. Zentgraf*, Y. Liu*, **M. H. Mikkelsen***, J. Valentine & X. Zhang. Plasmonic Luneburg and Eaton lenses. *Nature Nanotechnology* **6**, 151 (2011).
5. **M. H. Mikkelsen**, J. Berezovsky & D. D. Awschalom. Ultrafast optical manipulation of single electron spins in quantum dots. *Solid State Communications* **149**, 1451 (2009). *Invited article*.

4. J. Berezovsky*, **M. H. Mikkelsen***, N. G. Stoltz, L. A. Coldren & D. D. Awschalom. Picosecond coherent optical manipulation of a single electron spin in a quantum dot. *Science* **320**, 349–352 (2008). *Cover article*.
3. R. C. Myers, **M. H. Mikkelsen**, J.-M. Tang, A. C. Gossard, M. E. Flatté & D. D. Awschalom. Zero-field optical manipulation of magnetic ions in semiconductors. *Nature Materials* **7**, 203 (2008).
2. **M. H. Mikkelsen**, J. Berezovsky, N. G. Stoltz, L. A. Coldren & D. D. Awschalom. Optically detected coherent spin dynamics of a single electron in a quantum dot. *Nature Physics* **3**, 770 (2007).
1. J. Berezovsky, **M. H. Mikkelsen**, O. Gywat, N. G. Stoltz, L. A. Coldren & D. D. Awschalom. Nondestructive optical measurements of a single electron spin in a quantum dot. *Science* **314**, 1916 (2006).

BOOK CHAPTERS

G. M. Akselrod & **M. H. Mikkelsen**. Controlled Radiative Dynamics Using Plasmonic Nanocavities. In *Handbook of Metamaterials and Plasmonics*, O. Hess and S. A. Maier, eds. World Scientific (2017).

G. M. Akselrod, D. R. Smith & **M. H. Mikkelsen**. Controlled and enhanced fluorescence using plasmonic nanocavities. In *Surface Plasmon Enhanced, Coupled and Controlled Fluorescence*, C. D. Geddes, ed. Wiley (2017).

M. H. Mikkelsen, R. C. Myers, G. D. Fuchs & D. D. Awschalom. Single spin coherence in semiconductors. In *Spintronics*, Elsevier, pp. 1–44 (2008).

SELECTED PATENTS AND APPLICATIONS

M. H. Mikkelsen, E. Shin, R. Bangle & N. C. Wilson. Metasurface-enhanced thermal photodetector and methods thereof. U.S. application 63/749,926 (2025).

J. Stewart & M. H. Mikkelsen. High Speed and Spectrally Selective Pyroelectric Detectors with Plasmonic Structures and Methods of Making and Using Same. U.S. application 16/894,692 (2020).

M. H. Mikkelsen, D. R. Smith & G. M. Akselrod. Nanopatch Antennas and Related Methods for Tailoring the Properties of Optical Materials and Metasurfaces. PCT/US15/55033 (2015). Licensed.

ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

Editorial service

- 2022–2024 Associate Editor, *Optica*.
 2015–present Editorial Advisory Board, *ACS Photonics*.

Selected conference leadership and committees

- 2026–present Technical Committee, Plasmonics and Metamaterials, CLEO/Europe–EQEC.
 2022–present Symposium Co-chair, Molecular Devices, MRS Spring Meeting.
 2019–present Technical Committee, 2D Photonic Materials and Devices, SPIE Photonics West.
 2017–present Technical Committee, Quantum Nanophotonic Materials, Devices, and Systems, SPIE Optics & Photonics.
 2019 Scientific and Technical Program Committee, Surface Plasmon Photonics (SPP9).
 2018 Technical Committee, Nonlinear Nanophotonics, Plasmonics, and Metamaterials, OSA Nonlinear Photonics.

Award committees and peer review

Selection committees: SPIE Early Career Achievement Award (2018–2022); APS Maria Goeppert Mayer Award (2017–2018).

Proposal reviewer for NIH, NSF, the U.S. Department of Energy, AFOSR, the Army Research Office, Research Corporation for Science Advancement, and the Center for Integrated Nanotechnologies. Journal reviewer for *Science*, *Nature*, *Nature Photonics*, *Nature Materials*, *Nature Nanotechnology*, *PNAS*, and other journals.

Scientific input to U.S. Department of Defense and NSF discussions on nanomaterials and quantum information science, including invited strategy workshops and briefings (2014–2021).

MENTORING AND OUTREACH

Academic mentoring. Faculty Coordinator, Women in Physics, Duke (2012–2017); invited speaker, MIT Rising Stars workshop (2018); career and fellowship panels at Duke and Gordon Research Seminars (2015–2023).

STEM outreach. Organizer and instructor, FEMMES outreach program for girls in grades 4–6 (2014–2022); optics activities for primary school pupils in Durham (2024); mentoring meetings with women students during university visits (since 2014).

Selected trainee distinctions. Beckman Postdoctoral Fellowship: Rachel Bangle (2022). NSF Graduate Fellowships: Hengming Li (2021) and Daniela Cruz (2013). NDSEG Fellowship: Jon Stewart (2017). Intelligence Community Postdoctoral Research Fellowship: Gleb M. Akselrod (2014).

SELECTED INVITED AND PLENARY LECTURES

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| 2026 | Tailoring Local Fields for Ultrafast Single Photon Sources. Metamaterials Congress, New York, USA. |
| 2026 | Squeezing Light Fields for Ultrafast Detectors and Single Photon Sources. META, Dublin, Ireland. |
| 2025 | Quantum Nanophotonics: Ultrafast Single Photon Sources to Biosensing. European Quantum Technologies Conference, Copenhagen, Denmark. |
| 2025 | Plasmonic Nanocavities: From Telecom Single Photon Sources to Ultrafast Detectors. Gordon Research Conference on Plasmonically Powered Processes, Ventura, USA. |
| 2025 | Ultrafast Colloidal Quantum Dot Single Photon Sources in the Telecom Using Plasmonic Nanocavities. Photonics North, Ottawa, Canada. |
| 2024 | Quantum Photonics with Single Colloidal Nanoparticles. Gordon Research Conference on Noble Metal Nanoparticles, South Hadley, USA. |
| 2024 | Plasmonic Nanocavities for Ultrafast Single Photon Sources in the Telecom. CLEO, Charlotte, USA. |
| 2023 | Strong Coupling with Nanogap Cavities. Strong Coupling with Organic Molecules Conference, La Jolla, USA. |
| 2021 | Applications of Metasurfaces: From Multispectral Imaging to Optical Communications and Biosensing. META, Warsaw, Poland (virtual). Plenary lecture. |
| 2020 | Atomic-scale Engineering for Ultrafast Optoelectronics. ACS Photonics Young Investigator Award Lecture, ACS Global Webinar Series. |
| 2020 | Extreme Photonics with Nanogap Cavities. Max Planck Lecture, Max Planck Institute for the Structure and Dynamics of Matter. |
| 2019 | New Designer Materials: Sculpting Electromagnetic Fields on the Atomic Scale. Materials Science Seminar, Caltech, Pasadena, USA. |
| 2018 | Quantum Nanophotonics. Maria Goeppert Mayer Award Talk, APS DAMOP Meeting, Fort Lauderdale, USA. Plenary lecture. |
| 2018 | Designer Materials: Leveraging Ultrathin Metallic Junctions. Applied Physics Colloquium, Harvard University, Cambridge, USA. |
| 2018 | Extreme Nanophotonics from Ultrathin Metallic Junctions. Quantum Science and Technology Seminar, ETH Zürich, Switzerland. |